

Lecture 2: R Markdown, Version Control with Git(Hub), and Other Productivity Tools

James Sears*

AFRE 891/991 FS 26

Michigan State University

*Parts of these slides are adapted from [“Advanced Data Analytics”](#) by Nick Hagerty and [“Data Science for Economists”](#) by Grant McDermott.

Table of Contents

1. [Prologue](#)
2. [R Markdown](#)
3. [Version Control](#)
4. [GitHub Desktop](#)
5. [Other Tips and Productivity Tools](#)
6. [Not Covered: Git\(Hub\) + RStudio](#)
7. [Not Covered: Troubleshooting Git Credential Issues in RStudio](#)

Prologue

Prologue

Before we dive in, let's double check that we all have

- ☑ Installed [R](#).
- ☑ Installed [RStudio](#).
- ☑ Signed up for an account on [Github](#)
- ☑ Applied for a [Github Education](#) license
- ☑ Installed [Git](#) and [Github Desktop](#)
- ☑ Log into your Github account on Github Desktop

R Markdown

R Markdown

Before we dive into version control, let's chat about **R Markdown**.

R Markdown is a document type that allows for integration of R code and output into a Markdown document.

Resources:

- Website: rmarkdown.rstudio.com
- [R Markdown Cheatsheet](#)
- Book: [R Markdown: The Definitive Guide](#) (Yihui Xie, JJ Allaire, and Garrett Grolemund)

R Markdown

Before we dive into version control, let's chat about **R Markdown**.

R Markdown is a document type that allows for integration of R code and output into a Markdown document.

Other points:

- We'll be completing assignments using R Markdown.
- FWIW, my lecture slides and notes are all written in R Markdown too.
(E.g. This slide deck is built using the xaringan package with the metropolis theme.)

R Markdown: Getting Started

☒ Installed [R](#).

☒ Installed [RStudio](#).

☐ Add the `rmarkdown` package

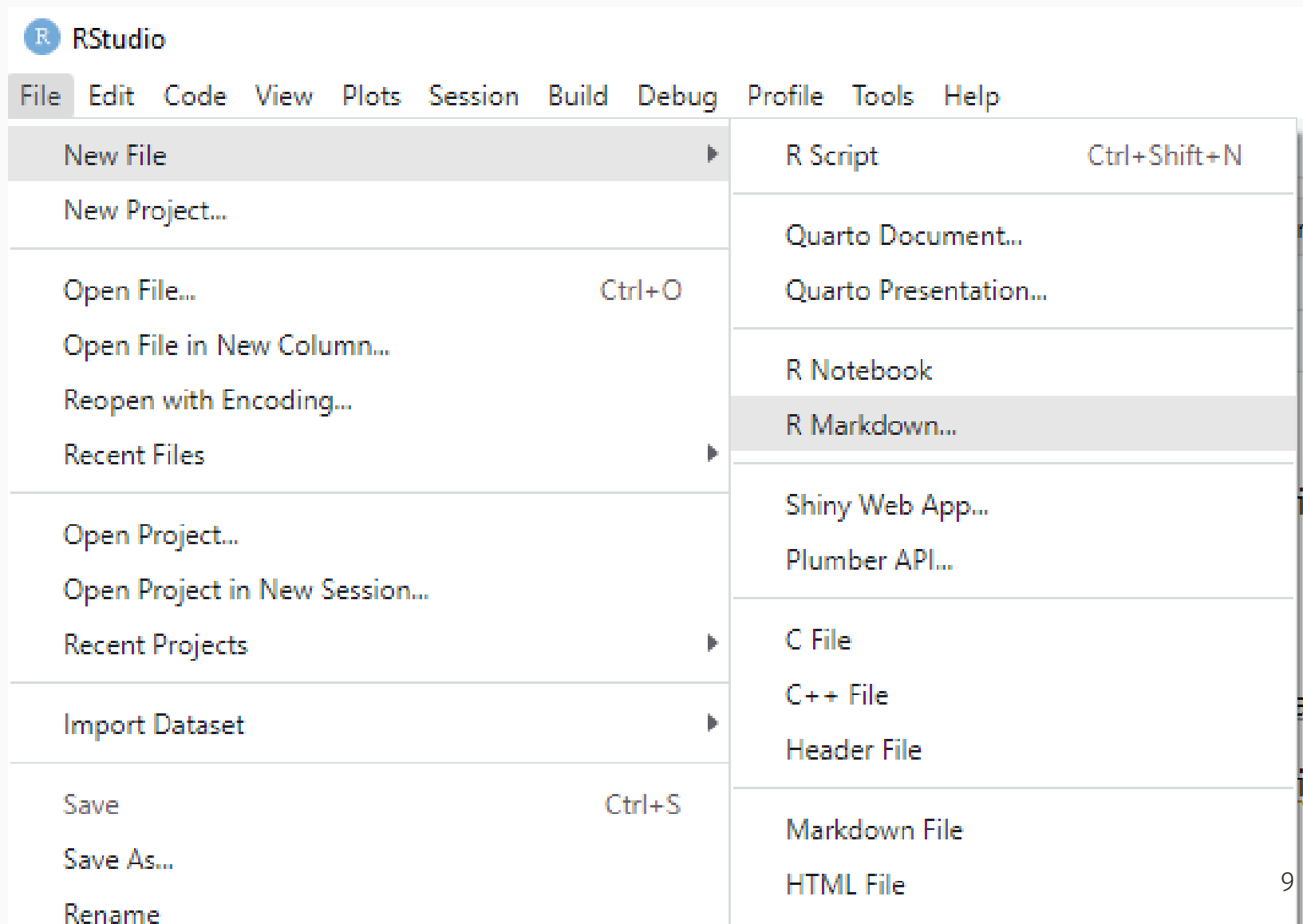
```
install.packages("rmarkdown")
```

☐ Install LaTeX

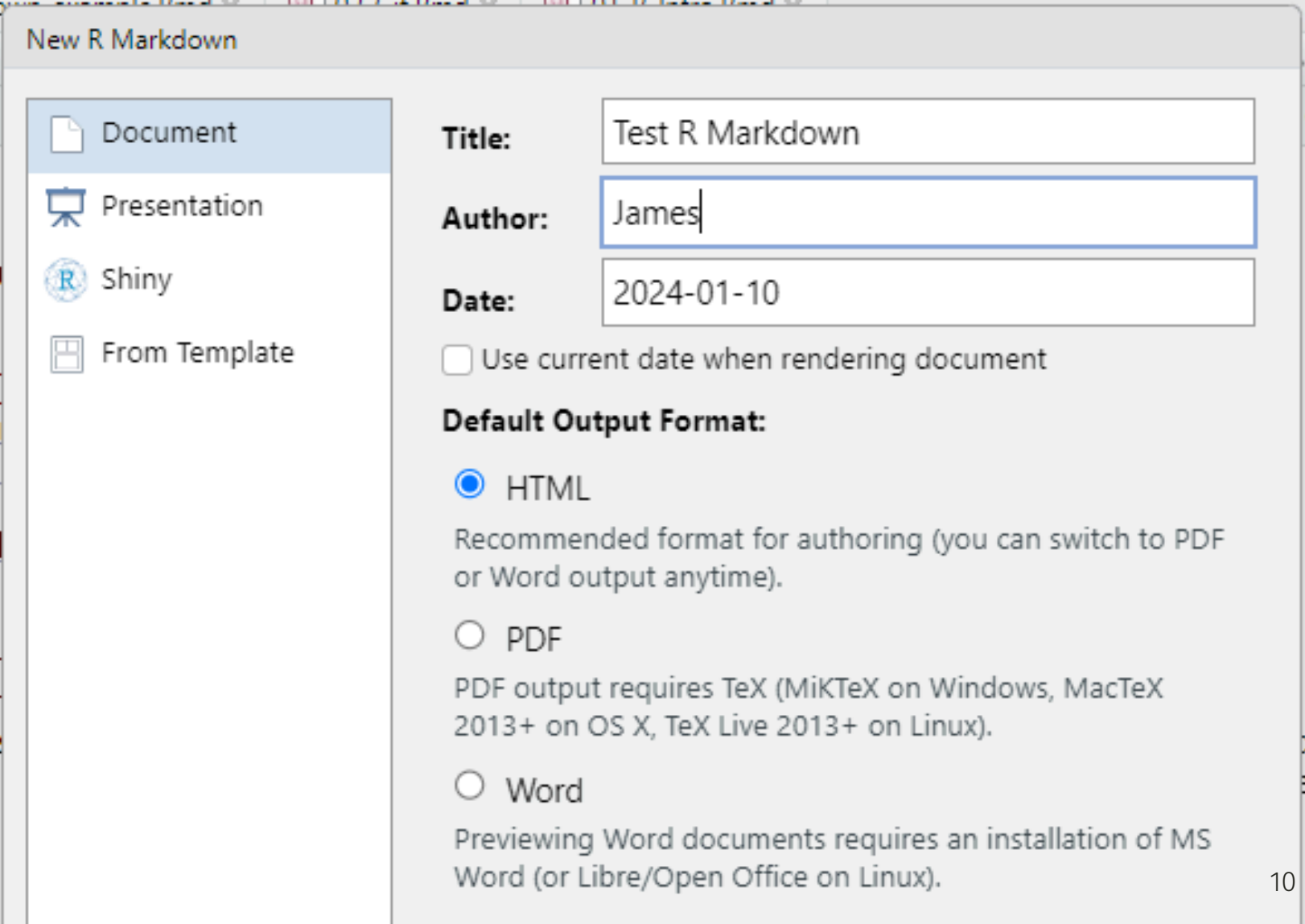
- If just for this, can use [TinyTex](#)

```
# Install only if you don't have LaTeX already  
install.packages("tinytex")  
tinytex::install_tinytex()
```


R Markdown: Creating a New .Rmd File



R Markdown: Creating a New .Rmd File



New R Markdown

 Document

 Presentation

 Shiny

 From Template

Title: Test R Markdown

Author: James

Date: 2024-01-10

☐ Use current date when rendering document

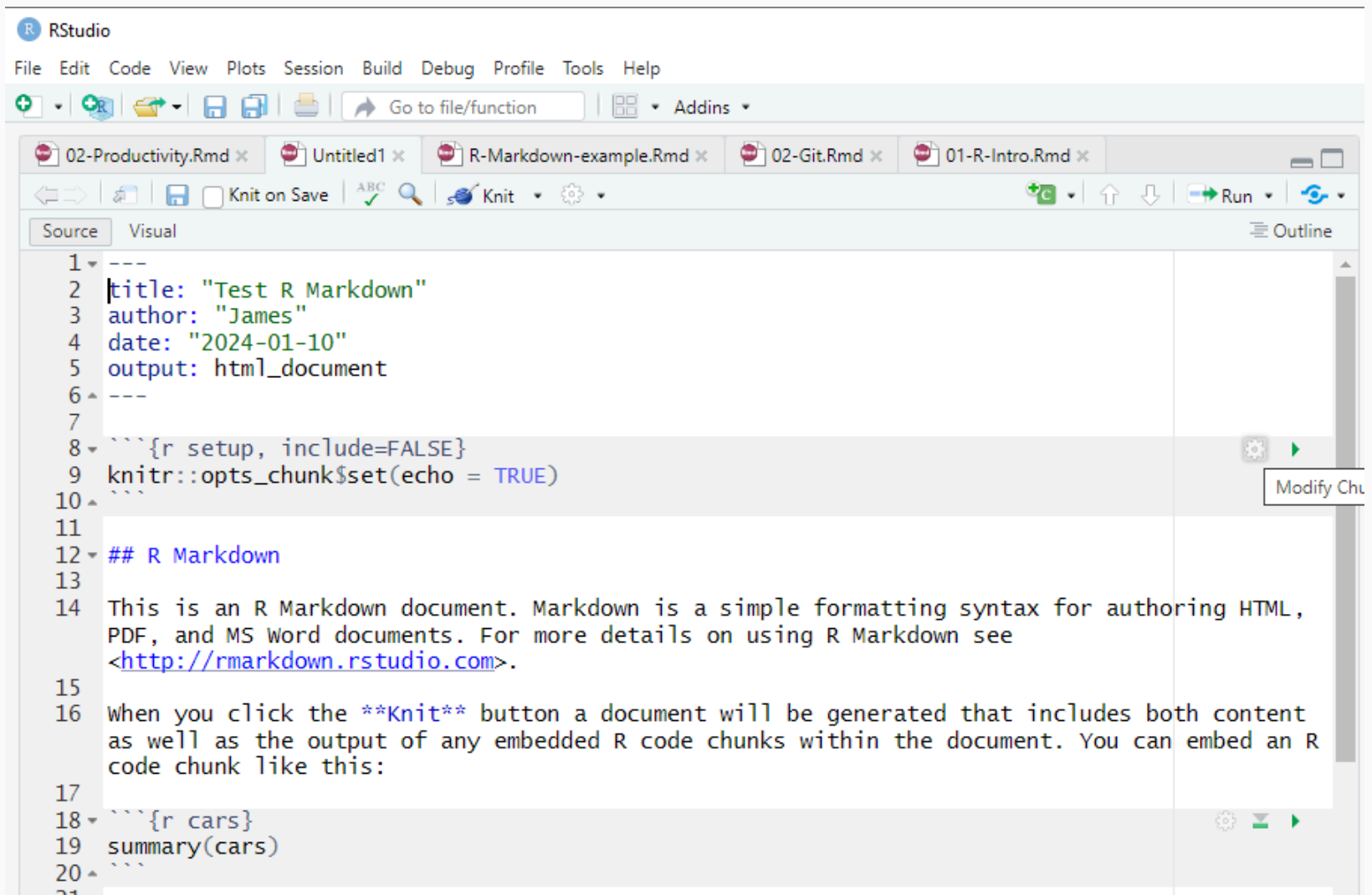
Default Output Format:

☒ HTML
Recommended format for authoring (you can switch to PDF or Word output anytime).

☐ PDF
PDF output requires TeX (MiKTeX on Windows, MacTeX 2013+ on OS X, TeX Live 2013+ on Linux).

☐ Word
Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux).

R Markdown: Creating a New .Rmd File



The screenshot shows the RStudio application window. The menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The toolbar contains icons for creating a new file, opening a file, saving, printing, and navigating. The file explorer shows several open files: 02-Productivity.Rmd, Untitled1, R-Markdown-example.Rmd, 02-Git.Rmd, and 01-R-Intro.Rmd. The editor is in the 'Source' view, showing the following R Markdown code:

```
1 ---
2 title: "Test R Markdown"
3 author: "James"
4 date: "2024-01-10"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML,
15 PDF, and MS Word documents. For more details on using R Markdown see
16 <http://rmarkdown.rstudio.com>.
17
18 When you click the Knit button a document will be generated that includes both content
19 as well as the output of any embedded R code chunks within the document. You can embed an R
20 code chunk like this:
21
22 ```{r cars}
23 summary(cars)
24 ```
```

A tooltip labeled 'Modify Chu' is visible over the chunk options icon (gear) next to the first code chunk.

R Markdown Components

R Markdown combines

1. **Markdown:** lightweight markup language
2. **LaTeX:** typesetting for math
3. **R:** include code and generate output

Let's do some practice: **open a new .Rmd file** and try adding content as we go

Markdown

Markdown allows for formatting text in a lightweight way

I highly recommend the handy [Markdown Guide](#) for more details

Markdown: Heading

Headings emphasize text and add chunks to your script

Largest heading with one leading # (slide title above)

Second Largest (##)

Third Largest (###)

Getting Smaller... (####)

Bold Text for comparison

Normal Text for comparison

Markdown: Text Format

Bold text with `**your text**`

Italicize with `*single asterisks*`

Add `code text` with grave accents (the back tick symbol, ```)

- The other output of the tilde key `~` on keyboard

End a line with two spaces to start a new paragraph

- or leave a line space between sentences

Can also start a new line with backslash (`\`)

Markdown: Text Format

Add superscripts² with ^carets^

Add ~~strikethroughs~~ with ~~double tildes~~

Add a line break (horizontal rule)

with ***

Markdown: Text Format

Draw **tables** using | and -

- | to delineate column breaks
- --- to separate headers from entries

```
| Col A | Col B | Col C |
```

```
| ---   | ---   | ---   |
```

```
| This  | is    | a      |
```

```
| Table |      | wow    |
```

Col A	Col B	Col C
This	is	a
Table		wow

Markdown: Text Format

You can adjust the **alignment** of table text by adding `:`'s in the second row:

- `:-----` for left-aligned
- `:---:---` for center-aligned
- `-----:` for right-aligned

Column A	Column B	Column C
Col A	is	left-aligned
Col B	is	center-aligned
Col C	is	right-aligned

Markdown: Lists

Add an **ordered list** with **1.**

1. First Item
2. Second Item
3. No need to change the number - keep using 1. It will automatically update.

Add an **unordered list** with *** or -**

- A thing
- Another related thing
 - Indent to nest
 1. Can mix ordered and unordered

Markdown: Inputs

Add a **link** with `[]()`

- `[text label](URL)`
- Add direct link with `<URL>` **<https://www.markdownguide.org>**

Add an image with `![]()`

- `![alt text](link)`
 - Where *link* is either a local file path or a URL

practice by adding `images/smile.png`:



Markdown: LaTeX

Another advantage of Markdown is that it integrates $\text{\LaTeX}$ functionality for typesetting math.

Add an **inline equation** with $\$TeX\$$

$$\textit{Var}(X) = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n} \quad Y_{it} = \beta_0 + \beta_1 X_{it} + \epsilon_{it}$$

Add multiple rows of LaTeX with

$$\$$$

LaTeX lines here

$$\$$$

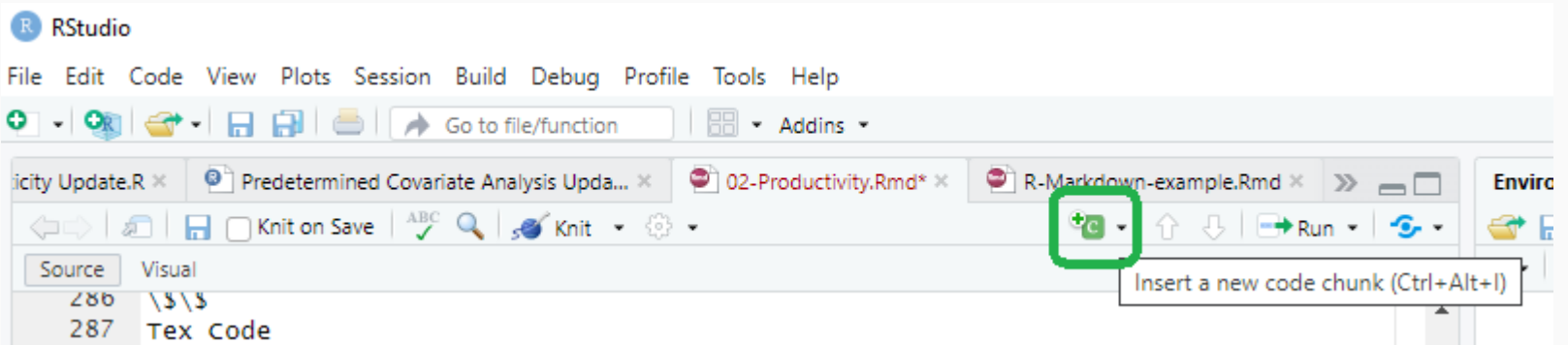
Use the **standard LaTeX commands** for symbols/characters

Rmd: R Code

R code is primarily executed with **code chunks**

Add a chunk with

- Cmd + Option + I (Ctrl + Alt + I on PC)
- The **Insert** button in the UI
- Manually type



Rmd: Code Chunks

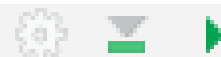
307

308 ~~~{r}

309

310 ~~~

311



Code chunks allow us to add as many lines of code as we want

- Output will appear underneath after executing the full chunk
- Can customize whether it runs, how output is displayed
- Can run manually
 - Line by line with `Cmd/Ctrl + Enter`
 - Entire chunk with `Run Entire Chunk` button

Code Chunk Options

You can **add chunk options** in brackets after `r` and separated by commas.

Some commonly-used options include:

Chunk label (i.e. `sum`)

- Must be unique names!
- Doesn't appear in compiled document

RMarkdown File

```
14 ~~~ {r sum}
15 2 + 2
16 ~~~
```

Compiled File

```
2 + 2
```

```
## [1] 4
```


Code Chunk Options

`echo = FALSE` will

- **run the code** and
- **display output** but
- **hide the code chunk** from the final document

RMarkdown File

```
19 {r, echo = FALSE}  
20 2 + 2  
21 }
```

Compiled File

```
## [1] 4
```

Code Chunk Options

`include = FALSE` will

- **run the code** but
- **hide the output** and
- **hide the code chunk** from the final document

RMarkdown File

```
26 {r, include = FALSE}  
27 2 + 2  
28 }
```

Compiled File

Code Chunk Options

You can **add chunk options** in brackets after `r` and separated by commas.

Some commonly-used options include:

`eval = FALSE` will

- **display code** in the chunk but
- **not run it**
- i.e. **no output** to display

RMarkdown File

```
33 {r, eval = F}  
34 2 + 2  
35
```

Compiled File

2 + 2

Code Chunk Options

You can also **change the color** of code chunks and/or output in the rendered document through code chunk options¹

- `class.source` to change the **code chunk**
- `output.source` to change the **output**

```
14 {r sum, class.source = "bg-success", output.source = "bg-danger"}  
15 2 + 2  
16
```

Code Chunk Options

You can also **change the color** of code chunks and/or output in the rendered document through code chunk options¹

- `class.source` to change the **code chunk**
- `output.source` to change the **output**

default

```
1 + 2 * 3 ^ 4
```

```
## [1] 163
```

bg-primary

```
1 + 2 * 3 ^ 4
```

```
## [1] 163
```

¹ We'll chat more about this when we get to web scraping, but what's actually happening here is that we're harnessing some built-in CSS classes to change the backgrounds. This also means that, using CSS, you can define custom classes and format things however you'd like.

Code Chunk Options

You can also **change the color** of code chunks and/or output in the rendered document through code chunk options¹

- `class.source` to change the **code chunk**
- `output.source` to change the **output**

bg-success

```
1 + 2 * 3 ^ 4
```

```
## [1] 163
```

bg-info

```
1 + 2 * 3 ^ 4
```

```
## [1] 163
```

¹ We'll chat more about this when we get to web scraping, but what's actually happening here is that we're harnessing some built-in CSS classes to change the backgrounds. This also means that, using CSS, you can define custom classes and format things however you'd like.

Code Chunk Options

You can also **change the color** of code chunks and/or output in the rendered document through code chunk options¹

- `class.source` to change the **code chunk**
- `output.source` to change the **output**

bg-warning

```
1 + 2 * 3 ^ 4
```

```
## [1] 163
```

bg-danger

```
1 + 2 * 3 ^ 4
```

```
## [1] 163
```

¹ We'll chat more about this when we get to web scraping, but what's actually happening here is that we're harnessing some built-in CSS classes to change the backgrounds. This also means that, using CSS, you can define custom classes and format things however you'd like.

Rmd: Inline Code

You can call R objects from earlier chunks **inline** with

```
- r -
```

```
four = 2+2
```

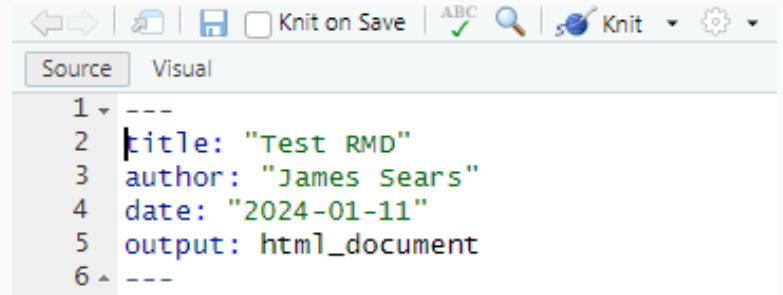
This can output in line with text: $2 + 2 = 4$

R Markdown File Organization

1. Header

RStudio automatically builds the R Markdown file from a template, which begins with a **header**

- Title
- Author
- Date
- Output Format
 - Main options²: HTML
(`html_document`), PDF
(`pdf_document`), LaTeX
(`latex_document`), or Word
(`word_document`)



The screenshot shows the RStudio Source editor with the 'Source' tab selected. The editor displays the header of an R Markdown file, which is enclosed in a dashed box. The header contains the following text: `---`, `title: "Test RMD"`, `author: "James Sears"`, `date: "2024-01-11"`, `output: html_document`, and `---`. The line numbers 1 through 6 are visible on the left side of the editor.

² See [CH 3 of "R Markdown: The Definitive Guide" for more on how to customize output formats](#)

2. R Setup

By default, RStudio adds a **setup** code chunk next.

```
8  ```{r setup, include=FALSE}
9  knitr::opts_chunk$set(echo = TRUE)
10 ```
11
```

- Can set global options
- Useful as your preamble
- For **R Notebooks**, this will automatically be run and is the only place where you can change your working directory

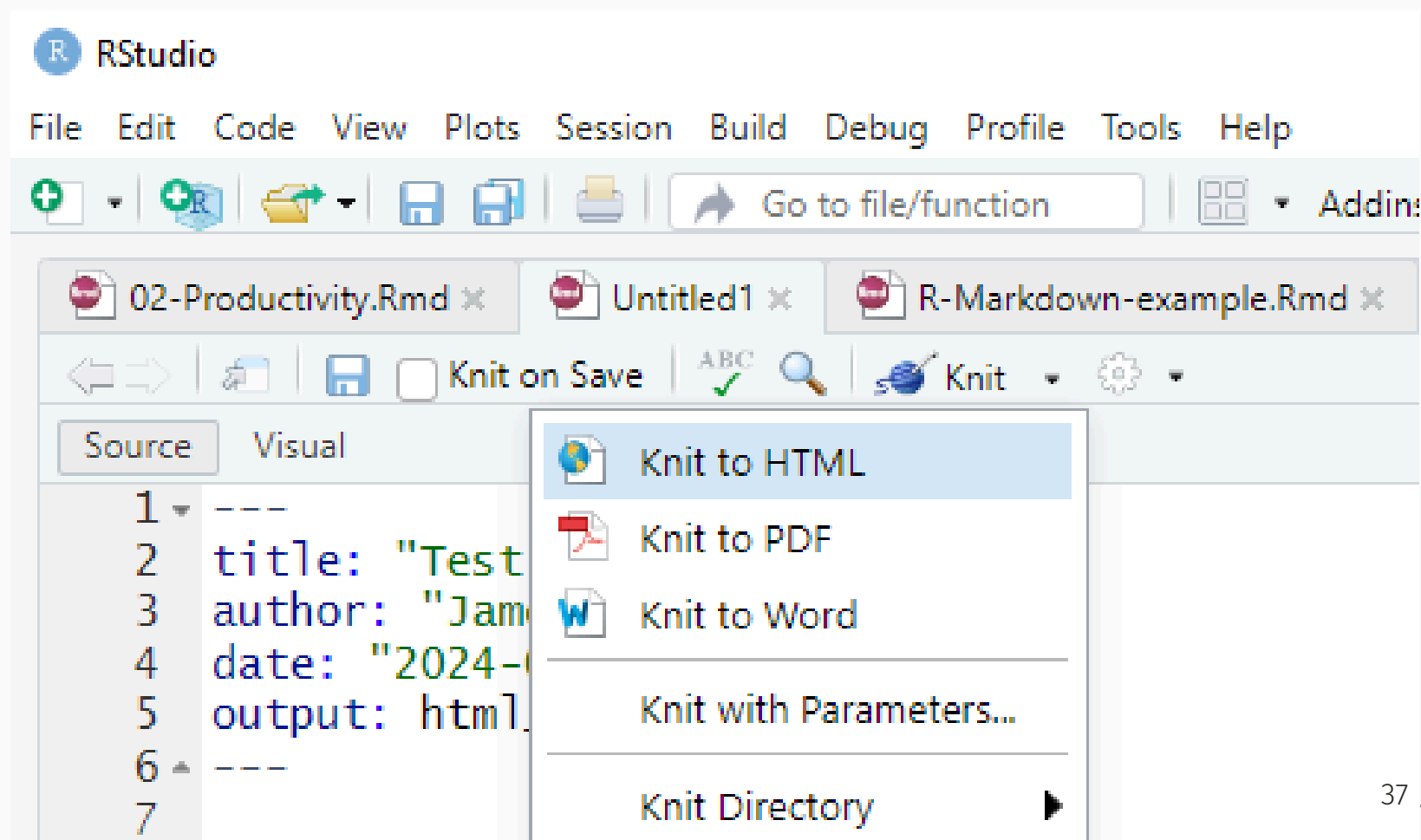
3. Contents

From here on you can build the report/notebook as needed for the task.

- Add any writing and outside graphics or **bibTeX citations**
- Add code chunks to carry out desired analysis
- Employ sections and formatting to structure the document as desired

Compiling/Knitting

When you are ready to compile your final document, use the **Knit** button or **Ctrl/Cmd + Shift + K**



R Markdown: Knit to Compile Output



The screenshot shows a web browser window displaying an R Markdown document. The browser's address bar shows the file path: `F:/OneDrive - Michigan State University/Teaching/MSU 2023-2024/AFRE 891 SS23/Lecture-Slides/02-Productivity/output/R-Markdown-example.html`. The browser's tab is labeled `R-Markdown-example.html`. The document content includes a title, author, date, a section header, an introductory paragraph, a paragraph about the **Knit** button, and a code chunk for `summary(cars)` with its corresponding output table.

Test R Markdown

James
2024-01-10

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

##	speed	dist
## Min.	: 4.0	Min. : 2.00
## 1st Qu.:	12.0	1st Qu.: 26.00
## Median :	15.0	Median : 36.00
## Mean :	15.4	Mean : 42.98
## 3rd Qu.:	19.0	3rd Qu.: 56.00
## Max.	:25.0	Max. :120.00

Markdown Practice!

Markdown Practice

1. Create a new R Markdown file named "R-Markdown-Ex.RMD"
2. In the setup chunk, load the `dslabs` and `tidyverse` packages
 - Use the `data()` function to read in the `divorce_margarine` dataset
3. Add a header labeled "Correlation vs. Causation" and a text explanation below for why we often want to differentiate between the two
4. Add a code chunk with the label `plot`
 - Type the following code:

```
ggplot(divorce_margarine) +  
  geom_point(aes(x = margarine_consumption_per_capita,  
                 y = divorce_rate_maine)) +  
  labs(title = "Relationship between Margarine Consumption and  
             Divorce Rates in Maine",  
        subtitle = "2000-2009",  
        x = "Margarine Consumption per Capita",  
        y = "Divorce Rate")
```

5. Knit and save a PDF/HTML copy of the file to the "output" folder

Table of Contents

1. **Prologue**
2. **R Markdown**
3. **Version Control**
4. **GitHub Desktop**
5. **Other Tips and Productivity Tools**
6. **Not Covered: Git(Hub) + RStudio**
7. **Not Covered: Troubleshooting Git Credential Issues in RStudio**